PF POST LAB TASK

Q1:

```
#include <stdio.h>

int main()

{

	for(int i=0;i<=8;i++)

	{

		for(int j=1;j<=i;j++)

		{

			if(j %4 ==0)

			{

				printf("0");

			} else {

				printf("1");

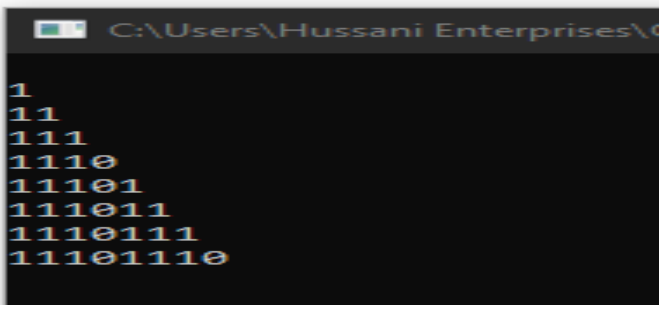
			}

		}

		printf("\n");

	}

}
```



Q2:

```
#include <stdio.h>

int main ()

{

	for(int i=1;i<=5;i++)

	{

		for(int j=1;j<=i;j++)

		{

			if(j %2==0)

			{

				printf("0");

			} else{

				printf("1");

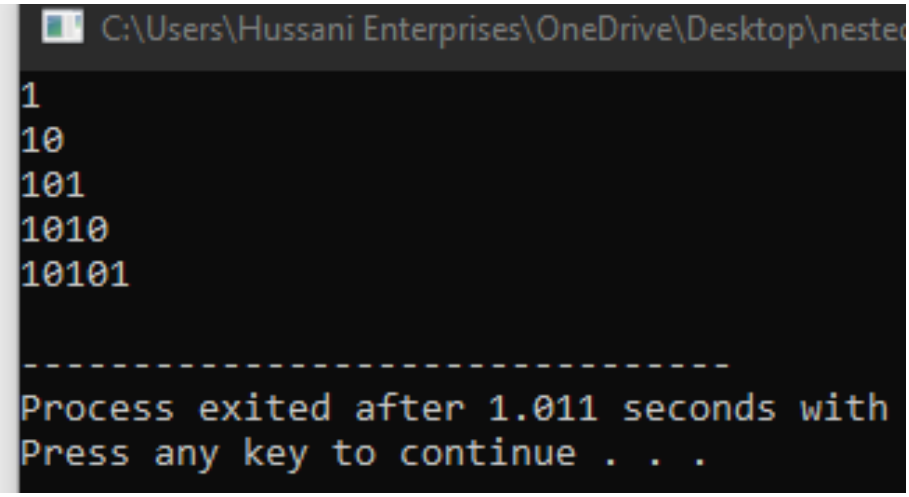
			}

		}

		printf("\n");

	}

}
```



Q3:

```
#include <stdio.h>

int main() {
   int a[3][3], b[3][3], sum[3][3];
   int i, j;

   printf("Enter elements of first matrix:\n");

   for (i = 0; i < 3; i++) {
      for (j = 0; j < 3; j++) {
         scanf("%d", &a[i][j]);
      }
   }

   printf("Enter elements of second matrix:\n");

   for (i = 0; i < 3; i++) {
      for (j = 0; j < 3; j++) {
         scanf("%d", &b[i][j]);
      }
   }

   // Add both matrices
```

```
for (i = 0; i < 3; i++) {
   for (j = 0; j < 3; j++) {
      sum[i][j] = a[i][j] + b[i][j];
   }
}

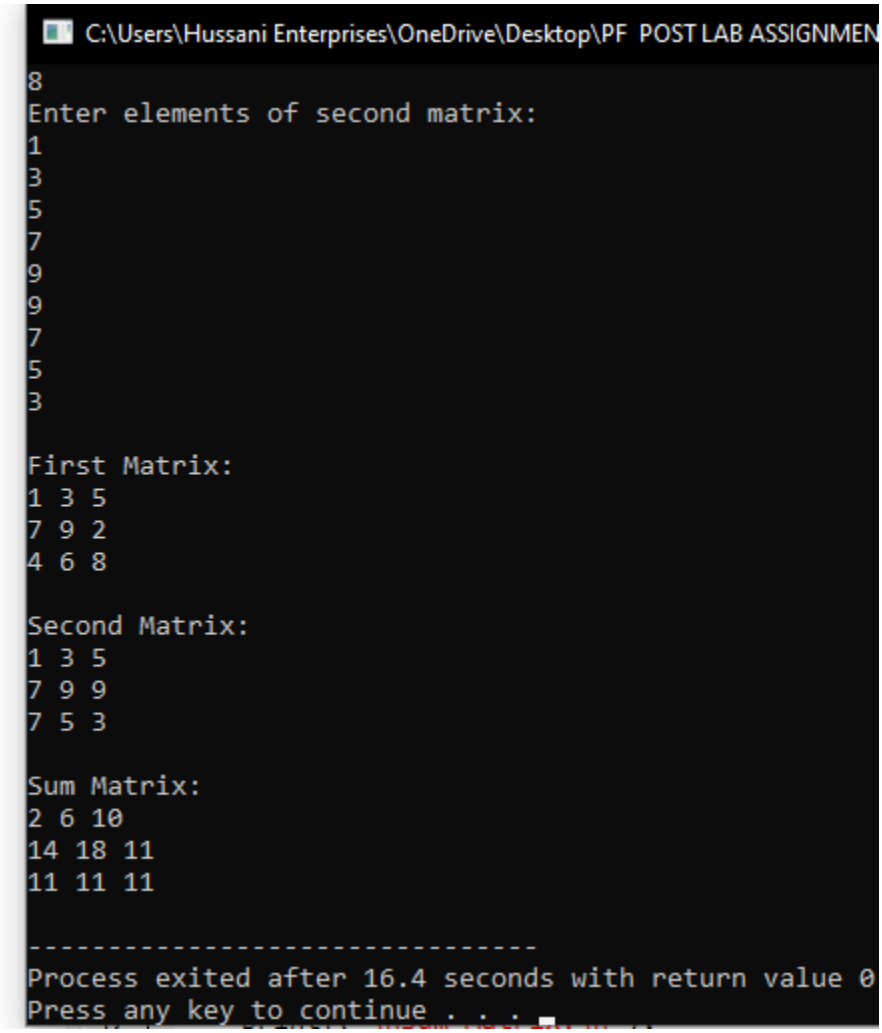
printf("\nFirst Matrix:\n");
for (i = 0; i < 3; i++) {
   for (j = 0; j < 3; j++) {
      printf("%d ", a[i][j]);
   }
   printf("\n");
}

printf("\nSecond Matrix:\n");
for (i = 0; i < 3; i++) {
   for (j = 0; j < 3; j++) {
      printf("%d ", b[i][j]);
   }
   printf("\n");
}

printf("\nSum Matrix:\n");
for (i = 0; i < 3; i++) {
   for (j = 0; j < 3; j++) {
      printf("%d ", sum[i][j]);
```

```
        }
        printf("\n");
    }

    return 0;
}
```



Q4:

```
#include <stdio.h>


int main() {
   // 2D Array: 4 students (rows), 2 exams per student (columns: 0 = Mid, 1 = Final)
   float marks[4][2];


   printf("=== Input Student Marks ===\n");
   // Input marks using nested loops
   for (int i = 0; i < 4; i++) {
      printf("Student %d:\n", i + 1);
      for (int j = 0; j < 2; j++) {
         if (j == 0) {
            printf("  Enter Mid marks: ");
         } else {
            printf("  Enter Final marks: ");
         }
         scanf("%f", &marks[i][j]);
      }
   }


   printf("\n========================================\n");
   printf("            RESULTS               \n");
   printf("========================================\n\n");
```

```
// 1. Calculate and display total marks for each student
printf("--- 1. Total Marks for Each Student ---\n");
for (int i = 0; i < 4; i++) {
   float total = marks[i][0] + marks[i][1];
   printf("Student %d Total Marks: %.2f\n", i + 1, total);
}
printf("\n");

// Initialize variables for highest and lowest tracking
float highest_mid = marks[0][0];
float highest_final = marks[0][1];
float highest_overall = marks[0][0];

float lowest_mid = marks[0][0];
float lowest_final = marks[0][1];
float lowest_overall = marks[0][0];

float sum_mid = 0.0;
float sum_final = 0.0;

// Loop to compute highest, lowest, and averages
for (int i = 0; i < 4; i++) {
   // Highest and Lowest in Mid
   if (marks[i][0] > highest_mid) highest_mid = marks[i][0];
   if (marks[i][0] < lowest_mid)  lowest_mid = marks[i][0];
```

```
    // Highest and Lowest in Final
    if (marks[i][1] > highest_final) highest_final = marks[i][1];
    if (marks[i][1] < lowest_final)  lowest_final = marks[i][1];

    // Highest and Lowest Overall across both exams
    for (int j = 0; j < 2; j++) {
        if (marks[i][j] > highest_overall) highest_overall = marks[i][j];
        if (marks[i][j] < lowest_overall)  lowest_overall = marks[i][j];
    }

    // Sums for Averages
    sum_mid += marks[i][0];
    sum_final += marks[i][1];
}

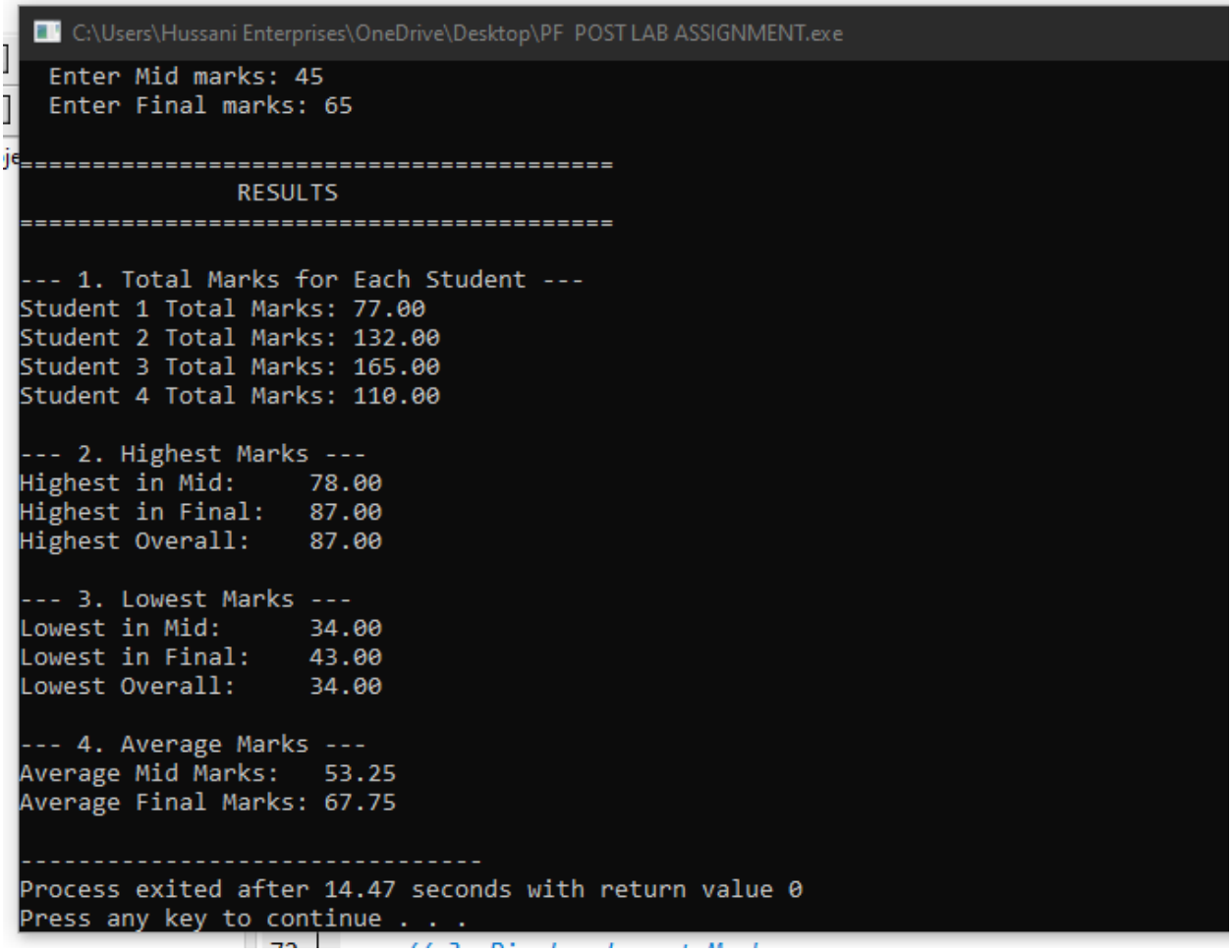
// 2. Display Highest Marks
printf("--- 2. Highest Marks ---\n");
printf("Highest in Mid:     %.2f\n", highest_mid);
printf("Highest in Final:   %.2f\n", highest_final);
printf("Highest Overall:    %.2f\n\n", highest_overall);

// 3. Display Lowest Marks
printf("--- 3. Lowest Marks ---\n");
printf("Lowest in Mid:      %.2f\n", lowest_mid);
printf("Lowest in Final:    %.2f\n", lowest_final);
printf("Lowest Overall:     %.2f\n\n", lowest_overall);
```

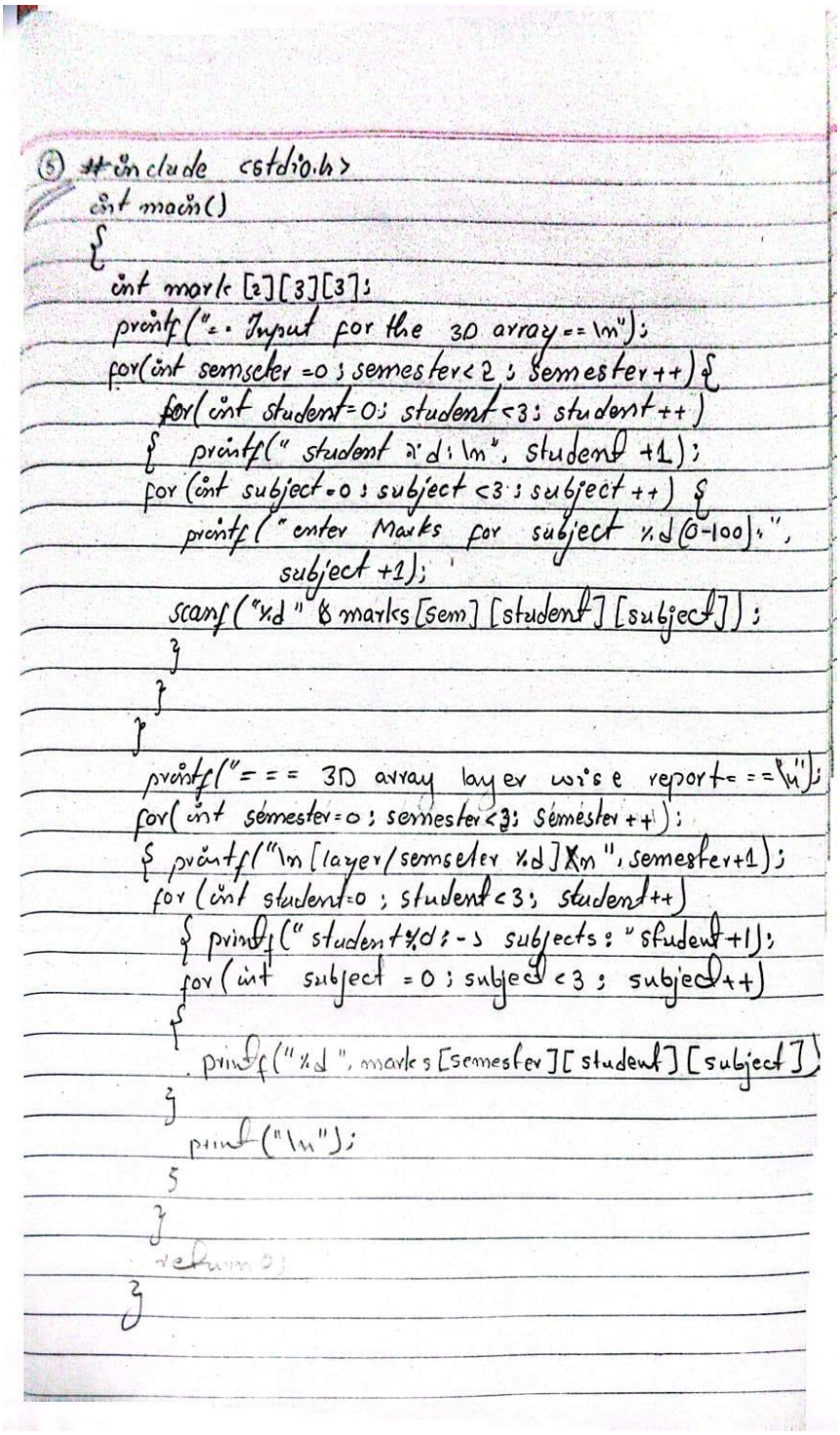
```
    // 4. Calculate and Display Averages
    float avg_mid = sum_mid / 4.0;
    float avg_final = sum_final / 4.0;

    printf("--- 4. Average Marks ---\n");
    printf("Average Mid Marks:   %.2f\n", avg_mid);
    printf("Average Final Marks: %.2f\n", avg_final);

    return 0;
}
```

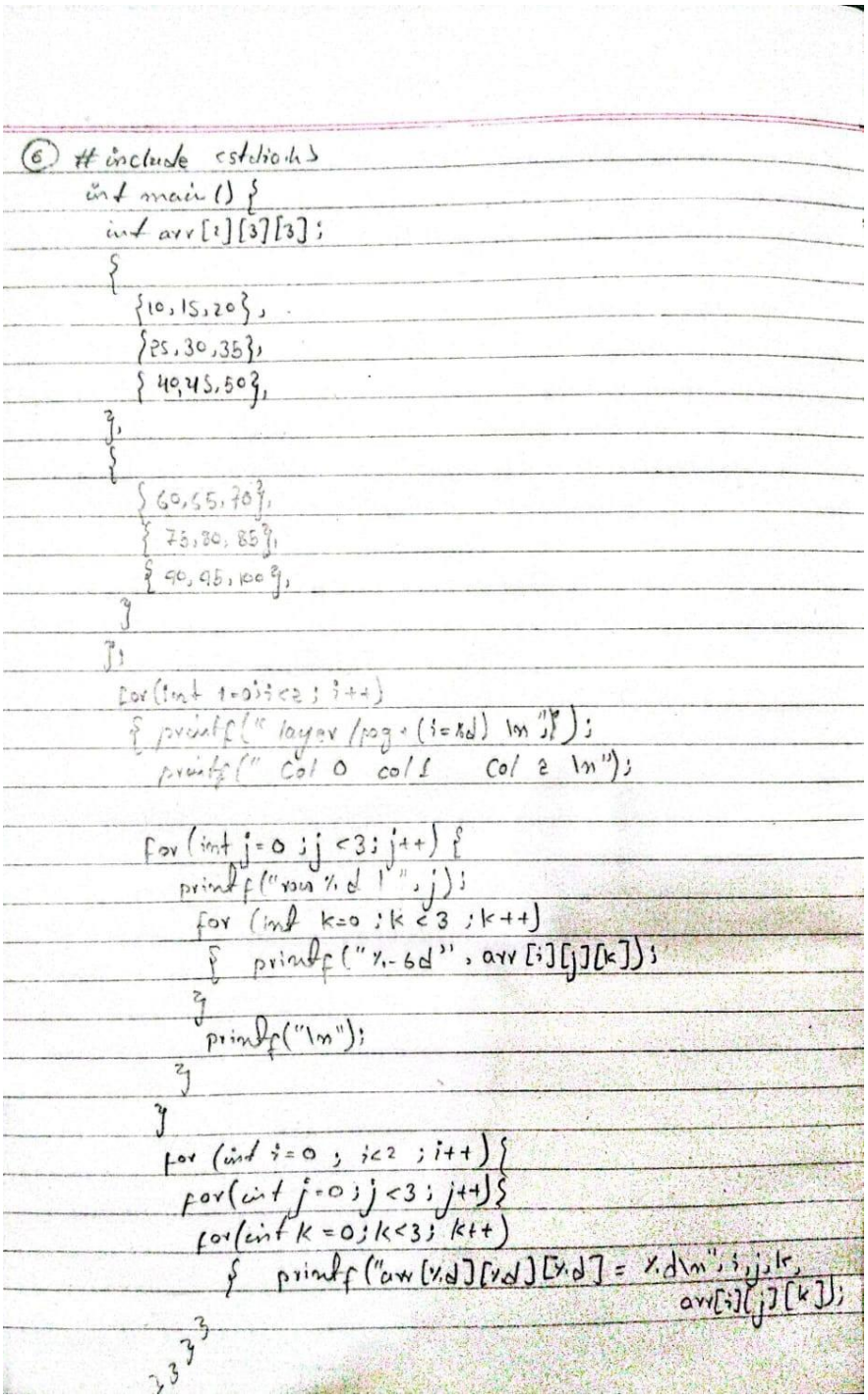


Q5;



```
⑤ #include <stdio.h>
int main()
{
 int marks[2][3][3];
 printf("=. Input for the 3D array ==\n");
 for(int semseter=0; semester<2; Semester++){
    for(int student=0; student<3; student++)
    { printf(" student %d: \n", student+1);
    for (int subject=0; subject<3; subject++) {
      printf("enter Marks for subject %d (0-100):",
             subject+1);
     scanf("%d" &marks[sem][student][subject]);
    }
   }
 }
 printf("=== 3D array layer wise report===\n");
 for(int semester=0; semester<3; Semester++);
 { printf("\n [layer/semseter %d]\n", semester+1);
  for (int student=0; student<3; student++)
    { printf("student%d:- subjects:" student+1);
    for(int subject=0; subject<3; subject++)
   {
     printf("%d ", marks[semester][student][subject]);
   }
    printf("\n");
  }
 }
 return 0;
}
```

Q6;



```
(6) #include <stdio.h>
int main () {
 int arr[2][3][3];
 {
   {10, 15, 20},
   {25, 30, 35},
   {40, 45, 50},
 },
 {
   {60, 65, 70},
   {75, 80, 85},
   {90, 95, 100},
 }
};
for(int i=0; i<2; i++)
 { printf(" layer /page = (i+1) \n");
   printf(" Col 0  col 1   Col 2 \n");

 for(int j=0; j<3; j++) {
   printf("row %d ", j);
    for (int k=0; k<3; k++)
     { printf("%-6d", arr[i][j][k]);
     }
     printf("\n");
   }
 }
 for (int i=0; i<2; i++){
  for(int j=0; j<3; j++){
  for(int k=0; k<3; k++)
    { printf("arr[%d][%d][%d] = %d\n", i, j, k,
                                    arr[i][j][k]);
}}}
```

Q7;

```
#include <stdio.h>


int main() {
   // 3D Array: 3 departments, 4 employees, 3 months[cite: 2]
   int salary[3][4][3];

   // Variables for highest salary tracking
   int max_salary = -1;
   int max_dept = 0, max_emp = 0, max_month = 0;

   printf("=== Input Employee Salaries ===\n");
   // Input salaries using nested loops
   for (int d = 0; d < 3; d++) {
      printf("\n--- Department %d ---\n", d + 1);
      for (int e = 0; e < 4; e++) {
         printf("  Employee %d:\n", e + 1);
         for (int m = 0; m < 3; m++) {
            printf("    Enter salary for Month %d: ", m + 1);
            scanf("%d", &salary[d][e][m]);

            // Track highest salary dynamically
            if (salary[d][e][m] > max_salary) {
               max_salary = salary[d][e][m];
               max_dept = d;
```

```
            max_emp = e;
            max_month = m;
        }
    }
  }
}

printf("\n=======================================\n");
printf("            RESULTS                \n");
printf("=======================================\n\n");

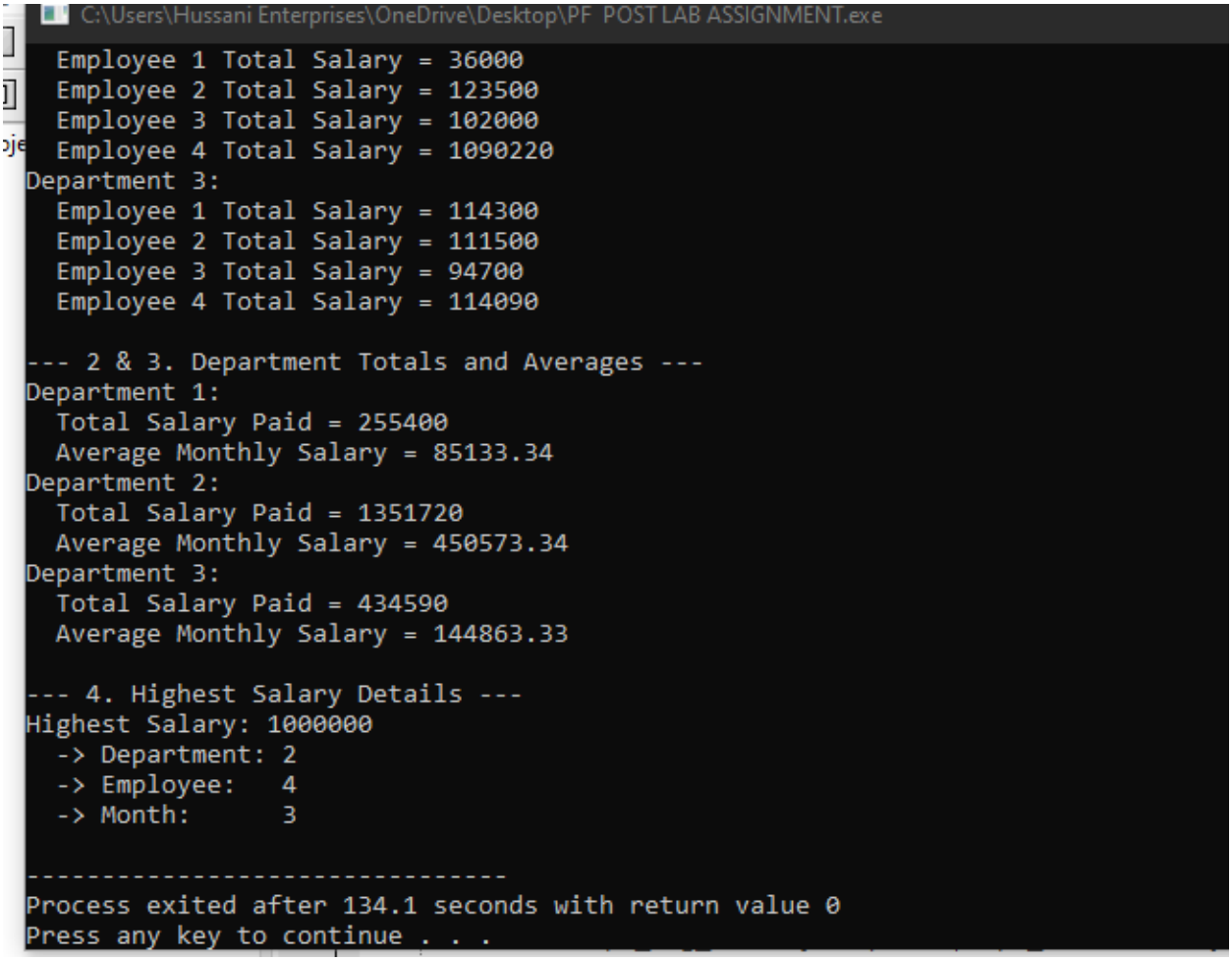
// 1. Total salary of each employee over the 3 months[cite: 2]
printf("--- 1. Total Salary of Each Employee (Over 3 Months) ---\n");
for (int d = 0; d < 3; d++) {
  printf("Department %d:\n", d + 1);
  for (int e = 0; e < 4; e++) {
    int emp_total = 0;
    for (int m = 0; m < 3; m++) {
      emp_total += salary[d][e][m];
    }
    printf("  Employee %d Total Salary = %d\n", e + 1, emp_total);
  }
}
printf("\n");
```

```
   // 2. Total salary paid by each department & 3. Average monthly salary of each department[cite: 2]
   printf("--- 2 & 3. Department Totals and Averages ---\n");
   for (int d = 0; d < 3; d++) {
      int dept_total = 0;
      for (int e = 0; e < 4; e++) {
         for (int m = 0; m < 3; m++) {
            dept_total += salary[d][e][m];
         }
      }
      // Average monthly salary of the department (total divided by 3 months)
      float dept_avg_monthly = (float)dept_total / 3.0;

      printf("Department %d:\n", d + 1);
      printf("  Total Salary Paid = %d\n", dept_total);
      printf("  Average Monthly Salary = %.2f\n", dept_avg_monthly);
   }
   printf("\n");

   // 4. Highest salary with its department, employee, and month[cite: 2]
   printf("--- 4. Highest Salary Details ---\n");
   printf("Highest Salary: %d\n", max_salary);
   printf("  -> Department: %d\n", max_dept + 1);
   printf("  -> Employee:   %d\n", max_emp + 1);
   printf("  -> Month:      %d\n", max_month + 1);
```

```
    return 0;
}
```


C:\Users\Hussani Enterprises\OneDrive\Desktop\PF POST LAB ASSIGNMENT.exe
Employee 1 Total Salary = 36000
Employee 2 Total Salary = 123500
Employee 3 Total Salary = 102000
Employee 4 Total Salary = 1090220
Department 3:
Employee 1 Total Salary = 114300
Employee 2 Total Salary = 111500
Employee 3 Total Salary = 94700
Employee 4 Total Salary = 114090
--- 2 & 3. Department Totals and Averages ---
Department 1:
Total Salary Paid = 255400
Average Monthly Salary = 85133.34
Department 2:
Total Salary Paid = 1351720
Average Monthly Salary = 450573.34
Department 3:
Total Salary Paid = 434590
Average Monthly Salary = 144863.33
--- 4. Highest Salary Details ---
Highest Salary: 1000000
-> Department: 2
-> Employee: 4
-> Month: 3
Process exited after 134.1 seconds with return value 0
Press any key to continue . . .

Q8;

```
#include <stdio.h>

int main() {
   // 3D Array: 2 sections, 5 students, 4 assignments[cite: 2]
   int marks[2][5][4];

   printf("=== Input Student Assignment Marks (0-10) ===\n");
   // Input marks using nested loops
   for (int s = 0; s < 2; s++) {
      printf("\n--- Section %d ---\n", s + 1);
      for (int st = 0; st < 5; st++) {
         printf("  Student %d:\n", st + 1);
         for (int a = 0; a < 4; a++) {
            printf("    Assignment %d Marks: ", a + 1);
            scanf("%d", &marks[s][st][a]);
         }
      }
   }

   printf("\n========================================\n");
   printf("            RESULTS                \n");
   printf("========================================\n\n");

   // 1. Total marks of each student over the 4 assignments[cite: 2]
   printf("--- 1. Total Marks of Each Student ---\n");
```

```
// We will store student totals to use later for highest calculation and threshold checks
int student_totals[2][5];

for (int s = 0; s < 2; s++) {
   printf("Section %d:\n", s + 1);
   for (int st = 0; st < 5; st++) {
      int total = 0;
      for (int a = 0; a < 4; a++) {
         total += marks[s][st][a];
      }
      student_totals[s][st] = total;
      printf("  Student %d Total = %d/40\n", st + 1, total);
   }
}
printf("\n");

// 2. Average marks of each assignment in each section[cite: 2]
printf("--- 2. Average Marks of Each Assignment per Section ---\n");
for (int s = 0; s < 2; s++) {
   printf("Section %d:\n", s + 1);
   for (int a = 0; a < 4; a++) {
      float sum_assignment = 0;
      for (int st = 0; st < 5; st++) {
         sum_assignment += marks[s][st][a];
      }
      float avg = sum_assignment / 5.0;
```

```
        printf("  Assignment %d Average = %.2f\n", a + 1, avg);
    }
}
printf("\n");

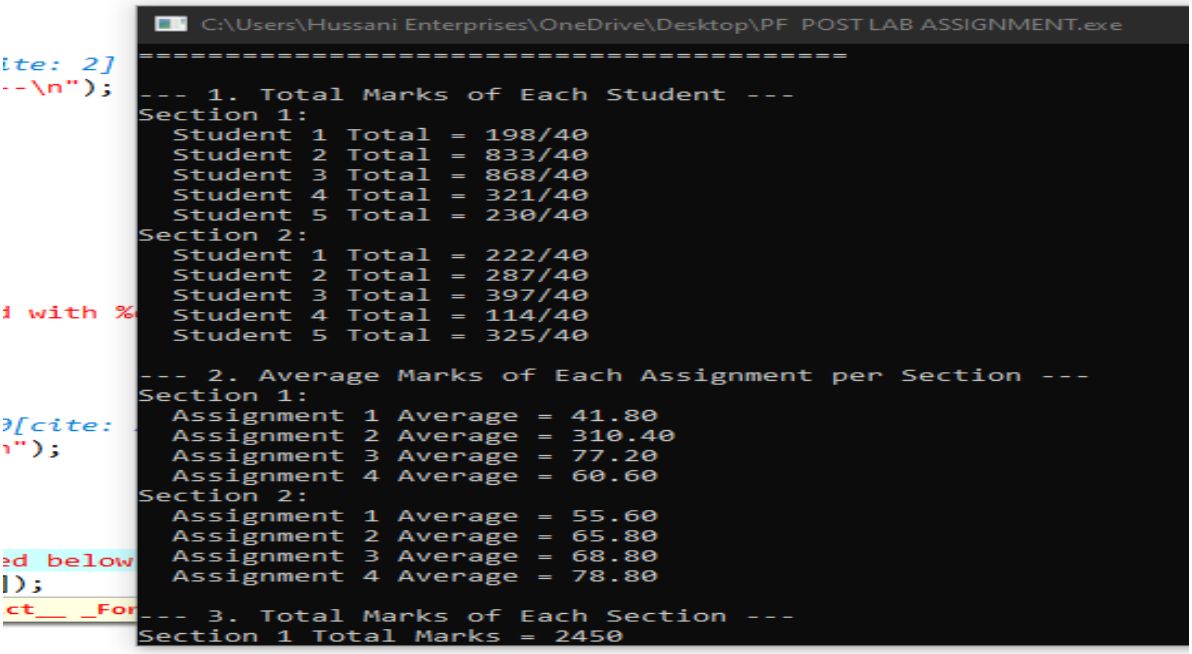
// 3. Total marks of each section[cite: 2]
printf("--- 3. Total Marks of Each Section ---\n");
for (int s = 0; s < 2; s++) {
    int section_total = 0;
    for (int st = 0; st < 5; st++) {
        for (int a = 0; a < 4; a++) {
            section_total += marks[s][st][a];
        }
    }
    printf("Section %d Total Marks = %d\n", s + 1, section_total);
}
printf("\n");

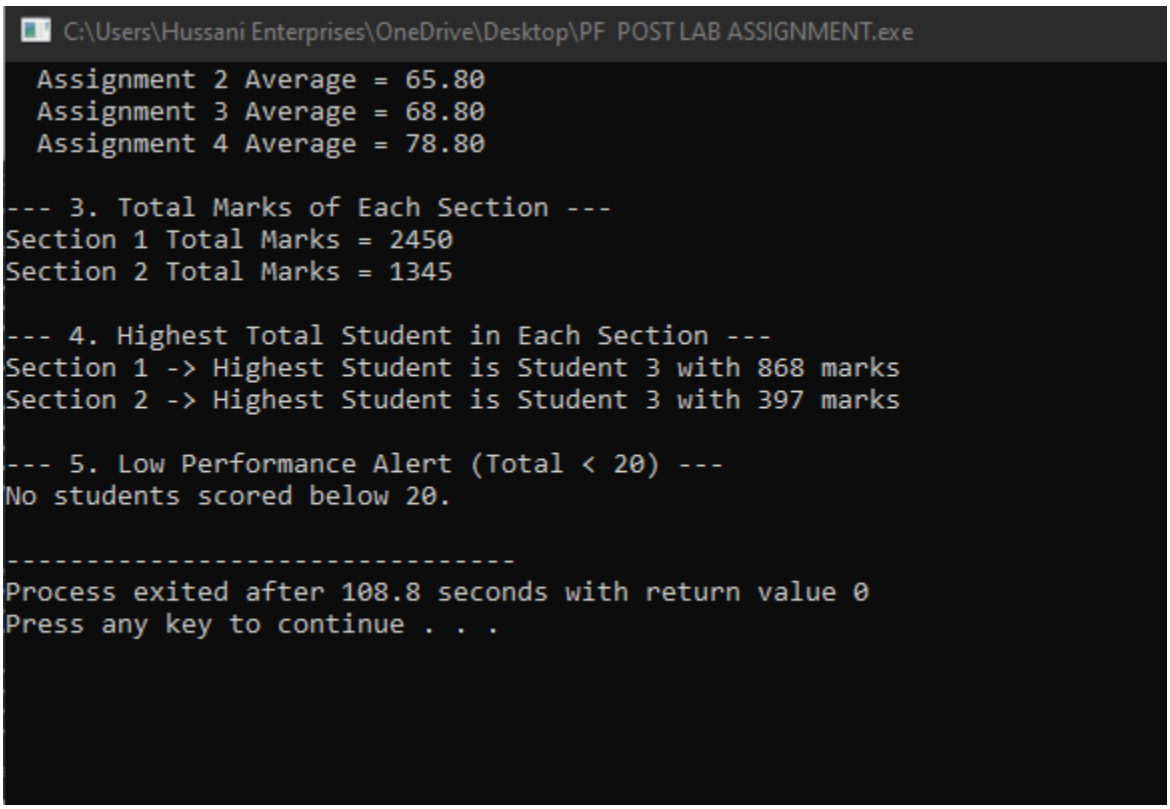
// 4. Student with the highest total in each section[cite: 2]
printf("--- 4. Highest Total Student in Each Section ---\n");
for (int s = 0; s < 2; s++) {
    int max_val = student_totals[s][0];
    int best_student = 0;
    for (int st = 1; st < 5; st++) {
        if (student_totals[s][st] > max_val) {
            max_val = student_totals[s][st];
```

```
            best_student = st;
        }
    }
    printf("Section %d -> Highest Student is Student %d with %d marks\n",
        s + 1, best_student + 1, max_val);
}
printf("\n");

// 5. Message for every student whose total is below 20[cite: 2]
printf("--- 5. Low Performance Alert (Total < 20) ---\n");
int low_count = 0;
for (int s = 0; s < 2; s++) {
    for (int st = 0; st < 5; st++) {
        if (student_totals[s][st] < 20) {
            printf("Alert: Section %d, Student %d scored below 20 (Total: %d)\n",
                s + 1, st + 1, student_totals[s][st]);
            low_count++;
        }
    }
}
if (low_count == 0) {
    printf("No students scored below 20.\n");
}

return 0;
}
```


C:\Users\Hussani Enterprises\OneDrive\Desktop\PF POST LAB ASSIGNMENT.exe
==========
--- 1. Total Marks of Each Student ---
Section 1:
Student 1 Total = 198/40
Student 2 Total = 833/40
Student 3 Total = 868/40
Student 4 Total = 321/40
Student 5 Total = 230/40
Section 2:
Student 1 Total = 222/40
Student 2 Total = 287/40
Student 3 Total = 397/40
Student 4 Total = 114/40
Student 5 Total = 325/40
--- 2. Average Marks of Each Assignment per Section ---
Section 1:
Assignment 1 Average = 41.80
Assignment 2 Average = 310.40
Assignment 3 Average = 77.20
Assignment 4 Average = 60.60
Section 2:
Assignment 1 Average = 55.60
Assignment 2 Average = 65.80
Assignment 3 Average = 68.80
Assignment 4 Average = 78.80
--- 3. Total Marks of Each Section ---
Section 1 Total Marks = 2450


C:\Users\Hussani Enterprises\OneDrive\Desktop\PF POST LAB ASSIGNMENT.exe
Assignment 2 Average = 65.80
Assignment 3 Average = 68.80
Assignment 4 Average = 78.80
--- 3. Total Marks of Each Section ---
Section 1 Total Marks = 2450
Section 2 Total Marks = 1345
--- 4. Highest Total Student in Each Section ---
Section 1 -> Highest Student is Student 3 with 868 marks
Section 2 -> Highest Student is Student 3 with 397 marks
--- 5. Low Performance Alert (Total < 20) ---
No students scored below 20.
Process exited after 108.8 seconds with return value 0
Press any key to continue . . .